

EXPERIMENT:9 FILE HANDLING IN C

Q1. Write a program to create a new file and write text into it.

Algorithm:

1* Start

2* Declare a file pointer FILE *fp.

3* Open a file in write mode using: fp = fopen("output.txt", "w");

4* If file opening fails (fp == NULL), show an error message and exit.

5* Ask the user to enter text.

6* Read the text from the user.

7* Write the text into the file using fprintf() or fputs().

8* Close the file using fclose(fp).

9* Display a message that the file has been written successfully.

10* Stop

Code:

```
#include <stdio.h>

int main() {
    FILE *fp;
    char text[200];

    // Open file in write mode
    fp = fopen("output.txt", "w");

    // Check if file opened successfully
    if (fp == NULL) {
        printf("Error! Could not create file.\n");
        return 1;
    }

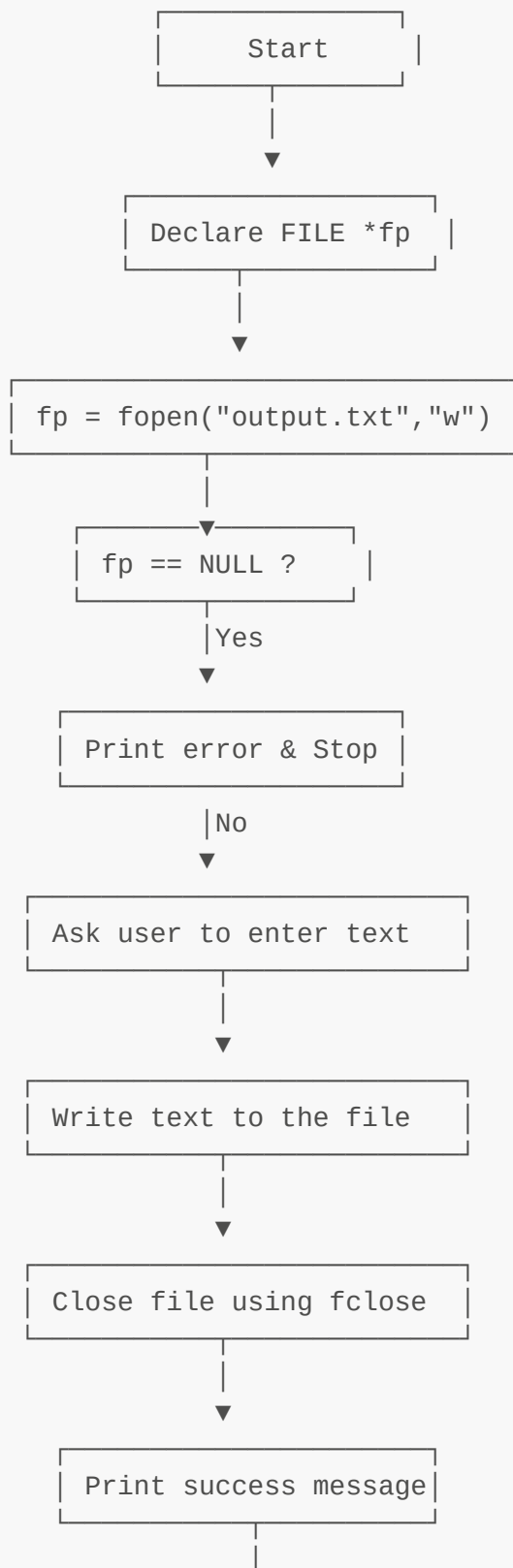
    // Get text from user
    printf("Enter text to write into the file:\n");
    fgets(text, sizeof(text), stdin);

    // Write text into file
    fprintf(fp, "%s", text);

    // Close file
    fclose(fp);
}
```

```
    printf("File written successfully!\n");  
  
    return 0;  
}
```

Flowchart:





Output:

```

cd "/home/aaditya/experiments-in-c/exp9/" && gcc exp9Q1.c -o exp9Q1 && "/home/aaditya/experiments-in-c/exp9/"exp9Q1
• aaditya@fedora:~/experiments-in-c$ cd "/home/aaditya/experiments-in-c/exp9/" && gcc exp9Q1.c -o exp9Q1 && "/home/aaditya/experiments-in-c/exp9/"exp9Q1
Enter text to write into the file:
hello myself aditya
File written successfully!
• aaditya@fedora:~/experiments-in-c/exp9$ █
  
```

Q2.Open an existing file and read its content character by character and then close the file.

Flowchart:

- 1-Start
- 2-Declare a file pointer FILE *fp.
- 3-Open the file in read mode using: fp = fopen("input.txt", "r");
- 4-If the file fails to open (fp == NULL), display an error message and stop.
- 5-Declare a character variable ch.
- 6-Read a character from the file using: ch = fgetc(fp)
- 7-While ch != EOF
- 8-Display the character on the screen
- 9-Read the next character
- 10-After reading all characters, close the file using fclose(fp).
- 11-Stop

Code:

```

#include <stdio.h>

int main() {
    FILE *fp;
    char ch;

    // Open the file in read mode
  
```

```
fp = fopen("input.txt", "r");

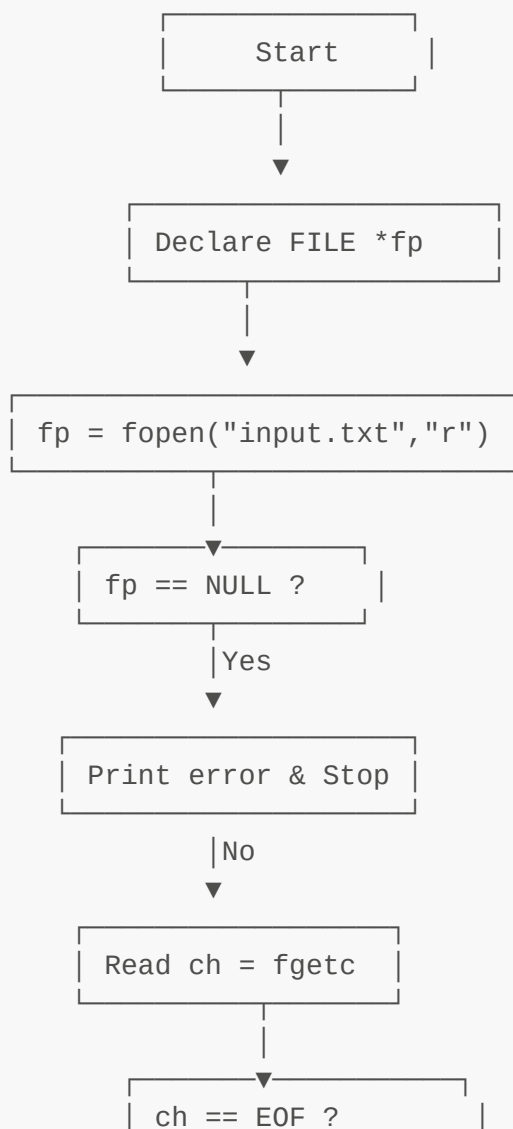
// Check if file opened successfully
if (fp == NULL) {
    printf("Error! Could not open file.\n");
    return 1;
}

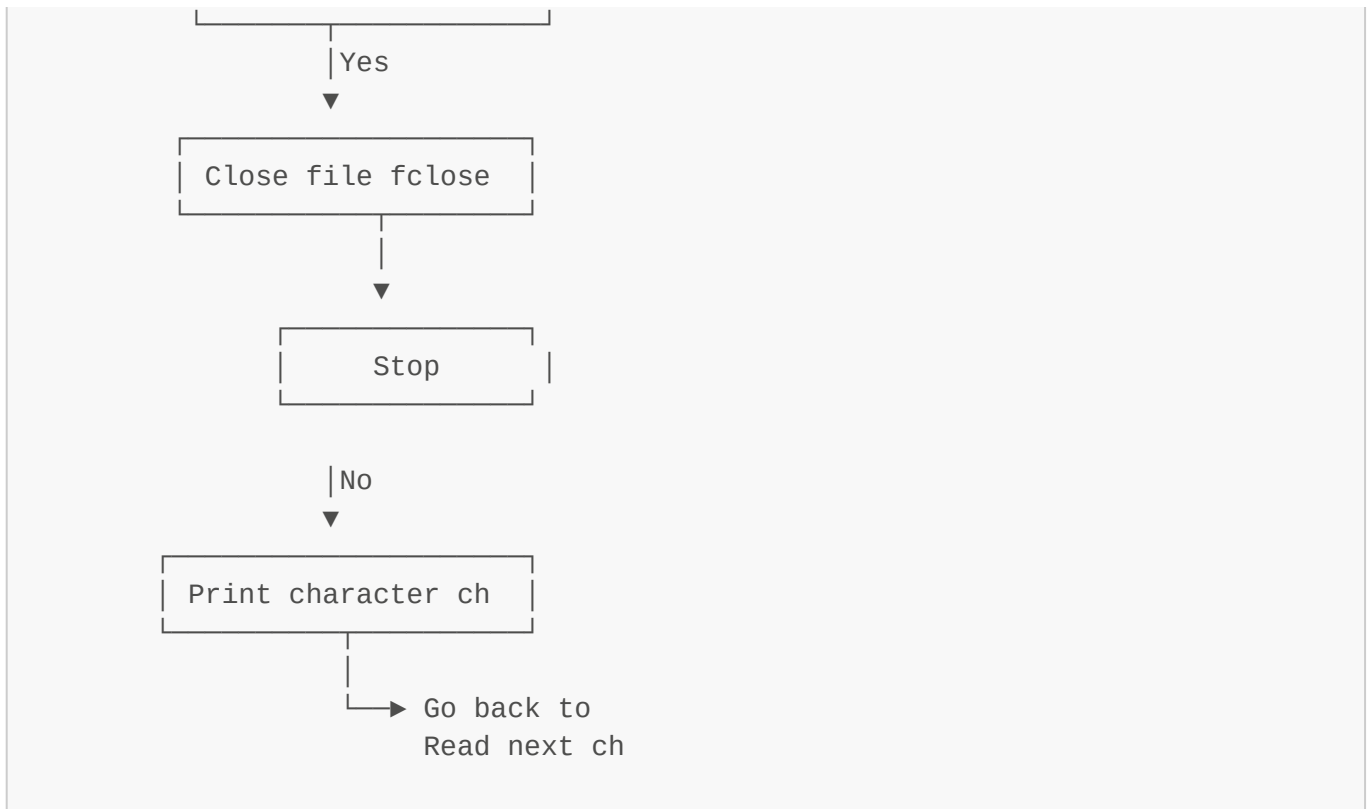
// Read and display characters until EOF
while ((ch = fgetc(fp)) != EOF) {
    putchar(ch);
}

// Close the file
fclose(fp);

return 0;
}
```

Flowchart:





Output:

```

cd "/home/aaditya/experiments-in-c/exp9/" && gcc exp9Q2.c -o exp9Q2 && "/home/aaditya/experiments-in-c/exp9/"exp9Q2
aaditya@fedora:~/experiments-in-c$ cd "/home/aaditya/experiments-in-c/exp9/" && gcc exp9Q2.c -o exp9Q2 && "/home/aaditya/experiments-in-c/exp9/"exp9Q2
hello myself aditya
aaditya@fedora:~/experiments-in-c/exp9$
  
```

Q3.Open a file read its content line by line and display each line on the console:

Algorithm:

- 1-Start
- 2-Declare a file pointer FILE *fp.
- 3-Declare a string buffer (e.g., char line[200] 😊).
- 4-Open the file in read mode using: fp = fopen("input.txt", "r");
- 5-If fp == NULL, print an error message and stop.
- 6-Read a line using: fgets(line, sizeof(line), fp)
- 7-While the line is not NULL:
- 8-Display the line on the console using printf("%s", line);
- 9-Read the next line using fgets()

10-Close the file using fclose(fp).

11-Stop

Code:

```
#include <stdio.h>

int main() {
    FILE *fp;
    char line[200];

    // Open file in read mode
    fp = fopen("input.txt", "r");

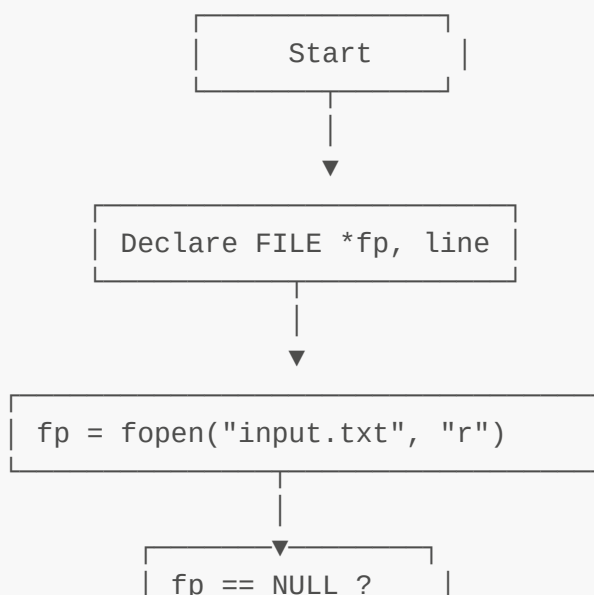
    if (fp == NULL) {
        printf("Error! Could not open file.\n");
        return 1;
    }

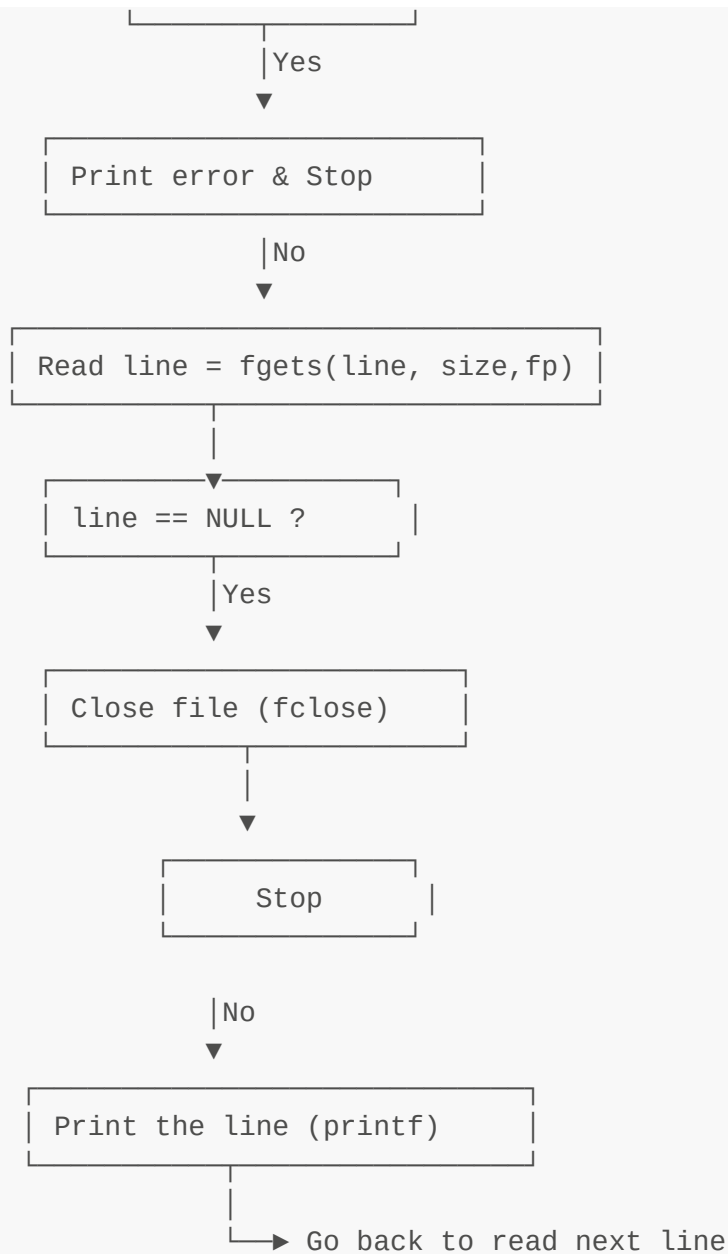
    // Read and display each line
    while (fgets(line, sizeof(line), fp) != NULL) {
        printf("%s", line);
    }

    // Close the file
    fclose(fp);

    return 0;
}
```

Flowchart:





Output:

```
cd "/home/aaditya/experiments-in-c/exp9/" && gcc exp9Q3.c -o exp9Q3 && "/home/aaditya/experiments-in-c/exp9/"exp9Q3
• aaditya@fedora:~/experiments-in-c$ cd "/home/aaditya/experiments-in-c/exp9/" && gcc exp9Q3.c -o exp9Q3 && "/home/aaditya/experiments-in-c/exp9/"exp9Q3
hello myself aditya
◦ aaditya@fedora:~/experiments-in-c/exp9$
```